

Systematic Mapping Research: Security in AI Agents

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A structured, rigorous method to survey and categorize existing literature on a topic. Originating from Systematic Literature Reviews (SLR) in medical research and formalized for software engineering by Petersen et al. (2008), systematic mapping takes a bird's-eye view — mapping *what* is being studied, *how much* research exists, and *where the gaps* are. Best suited for emerging or immature research areas.



Research Question

"What security threats, attack vectors, and defense mechanisms have been studied in relation to AI agents and the underlying AI/ML models that power them?"

This question is intentionally broad, designed to capture three key dimensions of the research landscape:

System-Level Security

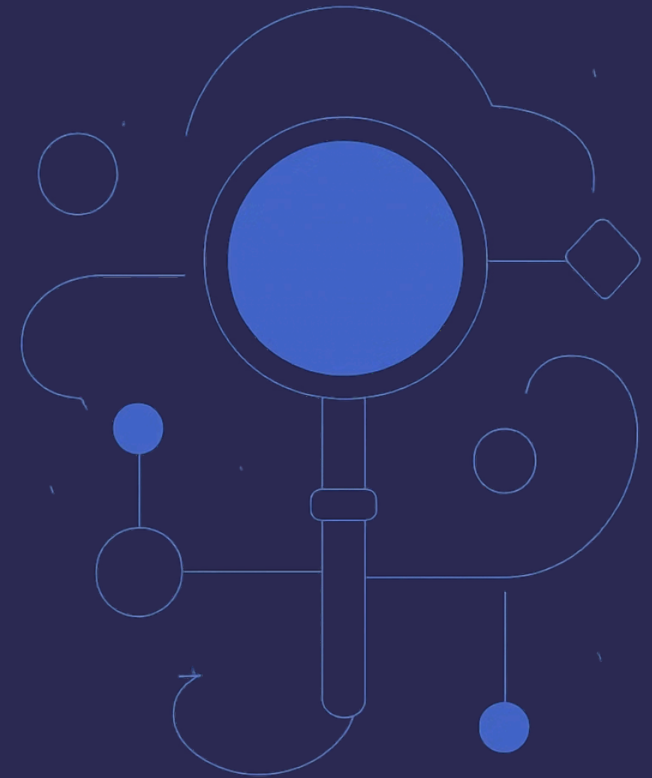
Security of agentic AI systems and architectures

Training-Time Attacks

Data poisoning and backdoor injection

Inference-Time Attacks

Adversarial evasion and model manipulation



Methodology: How the Search Was Done



Database

IEEE Xplore — peer-reviewed publications only

Time Window

2021–2026 (last 5 years)

Keywords

Agentic AI security, LLM agent vulnerabilities, prompt injection, data poisoning, backdoor attacks, adversarial examples, model evasion, federated learning attacks

Screening Results

52 candidate papers identified →
42 papers selected after filtering

Three Sub-Topics Identified

1

Sub-Topic1

Security Assessment of Agentic AI Systems

How are AI agents being attacked and defended at the system level?

2

Sub-Topic 2

Attacks on AI via Poisoning

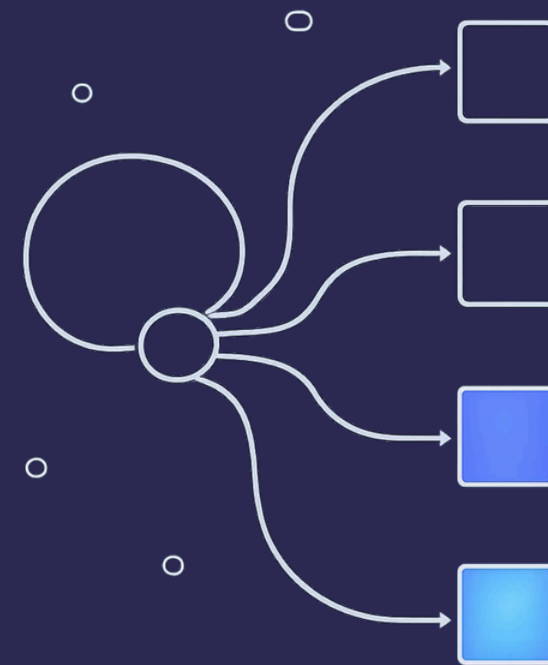
How is training data or model weights being manipulated?

3

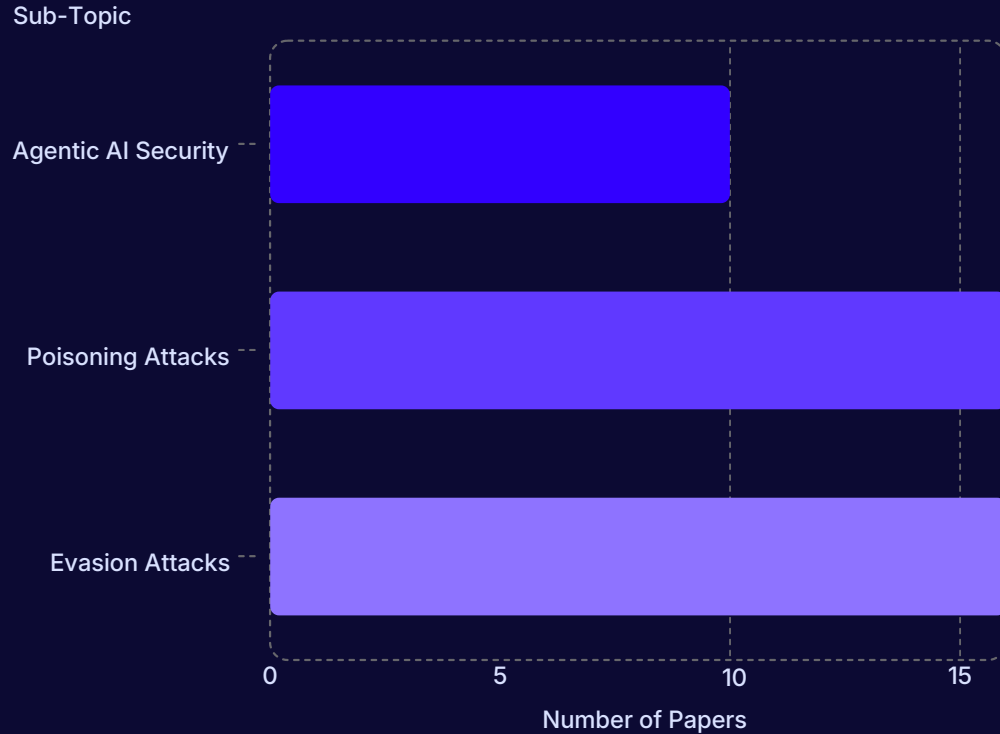
Sub-Topic 3

Attacks on AI Models via Evasion

How are inputs crafted to fool deployed models at inference time?



Paper Distribution Across Sub-Topics



What the Numbers Reveal

42 papers total were selected and classified across three sub-topics.

- Poisoning and Evasion each account for **16 papers** — reflecting more mature, established research streams
- Agentic AI Security has only **10 papers**, but the majority are from **2024–2026**, marking it as the newest and fastest-growing frontier

□ The imbalance signals an emerging research gap: classical adversarial ML techniques have not yet been systematically translated into the agentic AI context.

Agentic AI Security



Key Research Themes

- **MAAIS Framework** — multilayer security validated against MITRE ATLAS
- **Threat Taxonomy** — comprehensive classification of agentic AI threats and defenses
- **Prompt Injection Defense** — Polymorphic Prompt Assembling technique
- **RAG Security** — Retrieval-Augmented Generation as a core attack surface
- **RL-Based Defenders** — autonomous cybersecurity threat mitigation agents
- **LLM as Attacker** — security-augmented generative AI for cyberattack simulation

Poisoning Attacks

Attack Categories Covered

Classic Poisoning

Label flipping, instance injection, backdoor triggers

Clean-Label Attacks

Poison samples that appear legitimate to human reviewers

Federated Learning

Attacks on distributed training (FedCL, D2FL, FGLBA, FedRecAttack)

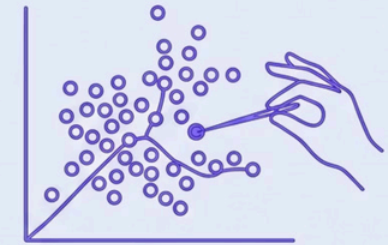
Domain-Specific

Speech recognition (SLU), IoT, recommender systems, model-weight poisoning



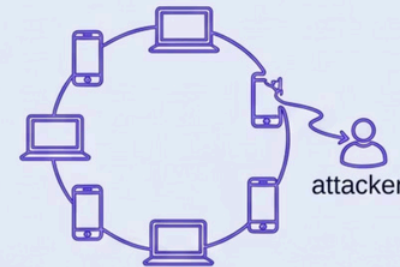
CLASSIC

Label Flipping
Instance Injection
Backdoors



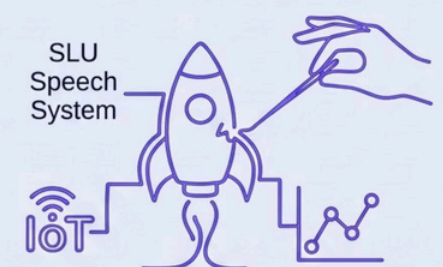
CLEAN-LABEL

Stealthy Poison Samples



FEDERATED LEARNING

Distributed Training Attacks
(FedCL, D2FL, FGLBA)



DOMAIN-SPECIFIC

SLU Speech, IoT, Recommender
Model-Weight Corruption

Evasion Attacks



Image Classification

FGSM, PGD, and BIM adversarial attack variants



Autonomous Driving

Digital and physical-world attacks on perception systems



Face & Audio

Adversarial patches, speaker recognition, and ASR evasion



Malware Detection

Adversarial binary obfuscation to evade classifiers



Text & NLP

Word manipulation and adversarial text generation



3D & Transferability

LiDAR-based attacks and black-box cross-model transfer (Foolmix)

Key Patterns Found

Poisoning is the most mature area

Covers training-time attacks across centralized, federated, graph-based, and recommender paradigms.

Evasion has diversified by domain

Each application 4 autonomous driving, IoT, NLP, audio, 3D 4 has its own sub-literature.

Agentic AI security is the newest frontier

Most papers are from 2024-2026, signaling rapid growth.

A critical gap exists at the intersection

Classical poisoning and evasion techniques have not yet been systematically translated into the agentic AI context.





Conclusion

Mapping Complete

42 IEEE papers (2021–2026) classified across 3 sub-topics, confirming Security in AI Agents as a legitimate and rapidly growing research domain.

Fragmentation Identified

Poisoning and evasion communities rarely overlap with agentic AI researchers — a clear structural gap in the literature.

Foundation Established

This systematic map provides a solid basis for targeted research and future systematic literature reviews.